

# VCP-Attack: Visual-Contrastive Projection for Transferable Black-Box Targeted Attacks on Large Vision-Language Models

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## Abstract

Large vision-language models (LVLMs) have achieved impressive performance across a variety of multimodal tasks, yet remain vulnerable to targeted adversarial attacks, particularly in black-box settings. In this paper, we propose **VCP-Attack**, a transferable targeted attack framework that combines structured contrastive supervision with subspace-guided perturbation optimization. Specifically, we employ a dynamic PCA-based projection to constrain perturbations within semantically meaningful low-dimensional subspaces, and design a multi-sample contrastive loss to align adversarial features with target semantics while pushing them away from the source semantics. Extensive experiments on seven open-source and three proprietary LVLMs—including GPT-4o, Claude, and Gemini—show that VCP-Attack achieves **state-of-the-art** performance in black-box targeted attacks. Under a fixed perturbation budget ( $\epsilon = 16/255$ ), our method achieves an average attack success rate (ASR) of 94.2% on open-source models and 83.1% on proprietary models, surpassing the strongest baselines by 23.3% and 16.8%, respectively. Notably, VCP-Attack achieves a 95.6% ASR on GPT-4o. Comprehensive ablation studies and visualizations further validate the effectiveness of the dynamic subspace projection and semantic contrastive supervision. While evaluated on image captioning, our approach is model-agnostic and exhibits strong potential for broader applications to black-box adversarial settings in vision-language tasks.

## 1. Introduction

Recent advances in high-performance computing and parallel processing have significantly accelerated the development of LVLMs, enabling strong performance across tasks such as image captioning [10, 11], visual question answering [18, 22], and multimodal reasoning [26, 28]. Among

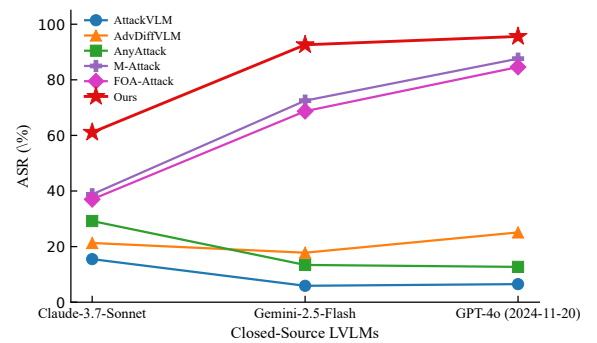


Figure 1. Our method achieves the highest ASR across proprietary LVLMs. Under a fixed perturbation budget ( $\epsilon = \frac{16}{255}$ ), our red-starred method outperforms all baselines on Claude 3.7, Gemini 2.5, and GPT-4o, highlighting its strong transferability and black-box effectiveness.

these, image captioning provides a structurally constrained yet semantically rich benchmark for analyzing adversarial behavior. Therefore, we adopt it as the primary evaluation task in this work. Modern LVLMs include open-source series like Qwen2.5-VL [25], Ovis-2 [21], and InternVL3 [5, 6], as well as commercial offerings such as GPT-4o [2], Claude-3.7 [1], and Gemini-2.5 [23]. Despite their impressive capabilities, these models remain highly vulnerable to adversarial prompts. Their deployment in real-world scenarios raises critical concerns around safety, bias, and misinformation, underscoring the need for rigorous evaluation and defense strategies. Among adversarial paradigms, targeted black-box transfer attacks are both realistic and challenging, as attackers aim to induce specific outputs from powerful LVLMs without accessing their internals. A standard approach is to perturb a surrogate model—typically a pretrained image or text encoder such as CLIP or BLIP—and transfer the crafted inputs to black-box targets. While this setup is widely adopted and also used in our work, existing methods often rely on point-wise cosine

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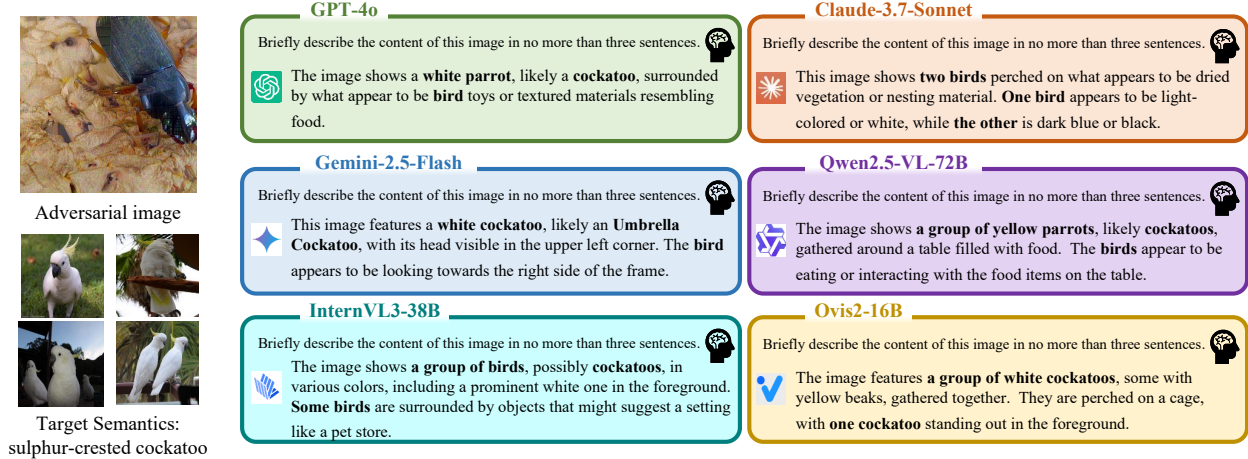


Figure 2. Targeted adversarial image crafted using VCP-Attack, along with textual descriptions generated by various commercial and open-source MLLMs in response to the prompt: “Briefly describe the content of this image in no more than three sentences.”

similarity in high-dimensional spaces, neglecting broader semantic structures. Such optimization yields perturbations that are noisy, redundant, or overfitted, limiting their generalization across architectures and prompts. Furthermore, these methods generally lack structured semantic guidance to influence the decision boundaries of LVLMs trained on rich visual-textual signals.

To address these challenges, we propose VCP-Attack, a Visual-Contrastive Projection attack that integrates contrastive supervision with subspace-constrained optimization. Instead of relying on a single reference embedding, VCP-Attack leverages multiple positive and negative samples to guide perturbations toward semantically coherent target clusters while repelling unrelated regions. To suppress noise and enhance generalization, all features are projected into a dynamically constructed PCA subspace, retaining only dominant semantic directions. This regularized contrastive objective encourages compact, transferable adversarial examples across diverse black-box LVLMs. Extensive experiments confirm that VCP-Attack transfers effectively to both open-source and proprietary LVLMs under strict black-box settings (Fig. 2).

The main contributions of this paper are as follows:

- **We propose VCP-Attack**, a novel targeted attack method that utilizes contrastive supervision with multiple positive and negative samples to align perturbations with target semantic regions while repelling unrelated concepts.
- **We introduce a PCA-based subspace optimization scheme**, which constrains perturbations to low-dimensional semantic directions, filtering out redundant gradients and enhancing semantic relevance and transferability.
- **We perform extensive black-box transfer experiments** on a variety of state-of-the-art LVLMs, demonstrating the

effectiveness and generalization of VCP-Attack across both open-source and commercial models.

## 2. Related Work

### 2.1. Transfer-Based Adversarial Attacks on Large Vision-Language Models

Transfer-based adversarial attacks are a widely adopted paradigm for evaluating LVLM robustness under black-box conditions.

AttackVLM [30] introduces a targeted black-box framework that aligns image features using surrogate vision encoders (e.g., CLIP, BLIP), showing strong transferability to MiniGPT-4 [31], LLaVA [20], and BLIP-2 [17]. It highlights the superiority of image-to-image matching over image-text objectives. Based on this paradigm, CWA [3] uses a calibration-weighted ensemble of multiple surrogates to enhance generalization, while SSA-CWA [8] further improves robustness via spectrum-based input transformations. Diffusion-based attacks such as AdvDiffuser [4] and AdvDiffVLM [12] generate unrestricted adversarial examples by perturbing latent denoising trajectories, though at the cost of visual fidelity. AnyAttack [29] learns a label-free adversarial generator via self-supervised pretraining on LAION-400M, enabling scalable and transferable attacks across VLMs. M-Attack [19] employs local-level cropping and alignment to inject semantic cues, achieving strong transferability to GPT-4o and Gemini. FOA-Attack [14] frames adversarial generation as a closed-form feature alignment problem over both patch and [CLS] token representations.

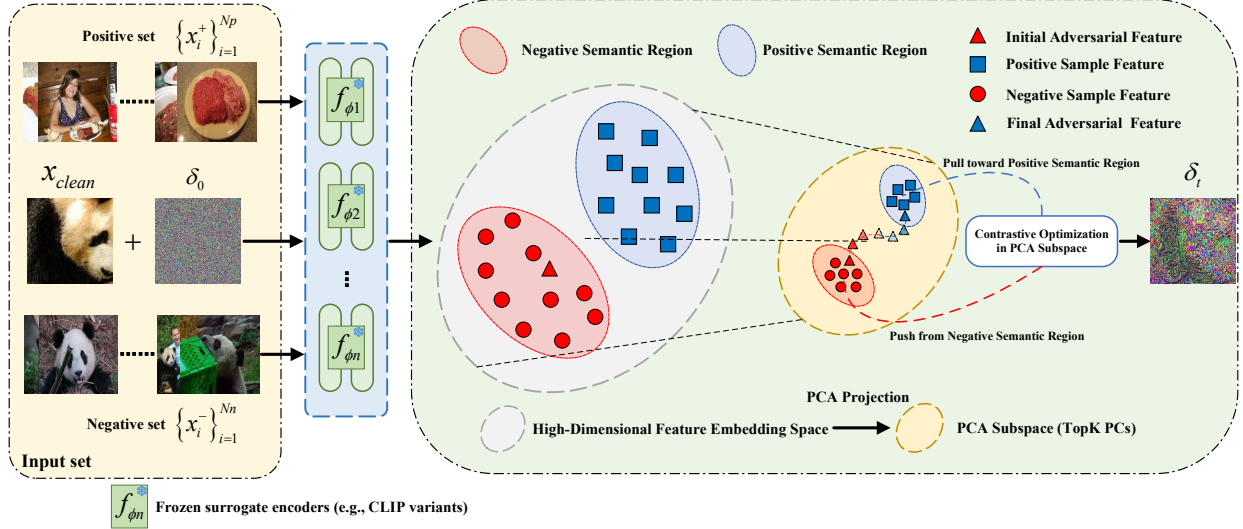


Figure 3. Overview of the proposed VCP-Attack framework. The method constructs a supervision set with positive and negative samples, extracts high-dimensional features via frozen encoders, and performs contrastive optimization in a PCA subspace to update adversarial perturbations.

## 2.2. Contrastive Learning for Adversarial Robustness

Contrastive learning has been widely explored to enhance adversarial robustness by stabilizing feature representations under perturbations. These methods conceptually inspire our contrastive supervision strategy.

RoCL [16] applies instance-level contrastive learning in adversarial training. ADVCL [9] improves adversarial robustness transfer from contrastive pretraining to finetuning by incorporating adversarial and high-frequency views, along with pseudo-supervised clustering. ACL [15] aligns representations under augmentations and adversarial noise to improve downstream robustness. FACTUAL [27] integrates supervised contrastive loss into adversarial training for SAR image classification using PGD and physical attacks (OTSA).

Contrastive learning has predominantly been employed to enhance model robustness in adversarial training or pre-training settings. In contrast, we explore its application on the offensive side—integrating contrastive supervision directly into the perturbation optimization process to generate semantically aligned and transferable adversarial examples.

## 3. Methodology

To facilitate understanding, Fig. 3 illustrates the overall framework of VCP-Attack, with key components detailed below.

### 3.1. Preliminary

We aim to perform targeted transfer-based attacks on LVLMs under black-box conditions. Given a clean image  $x_{\text{clean}} \in \mathbb{R}^D$  and a desired target semantics  $y_{\text{tar}}$ , the goal is to generate an adversarial example  $x_{\text{adv}} = x_{\text{clean}} + \delta$  such that the black-box LVLm  $M$  produces an output aligned with the semantics of  $y_{\text{tar}}$ :

$$M(x_{\text{adv}}, \text{Prompt}) \rightarrow y_{\text{tar}} \quad (1)$$

To enable such attacks without accessing  $M$ 's internal parameters or gradients, we rely on a set of frozen surrogate vision encoders  $\mathcal{F} = \{f_{\phi_1}, f_{\phi_2}, \dots, f_{\phi_t}\}$ ,  $f_{\phi} : \mathbb{R}^D \rightarrow \mathbb{R}^F$ , where each  $f_{\phi_i}$  maps an image to its high-dimensional feature embedding. We also construct a supervision set  $\mathcal{S} = \{\{x_i^+\}_{i=1}^{N_p}, \{x_j^-\}_{j=1}^{N_n}\}$ , where  $x_i^+$  are positive samples semantically aligned with the target semantics  $y_{\text{tar}}$ , and  $x_j^-$  are negative samples that share the original semantics of  $x_{\text{clean}}$ . The adversarial perturbation  $\delta$  is optimized to make the adversarial example  $x_{\text{adv}}$  align with the target semantics, while remaining imperceptible. It can be formulated as a constrained optimization problem:

$$\min_{\|\delta\|_{\infty} \leq \epsilon} \sum_{i=1}^t \mathcal{L}(f_{\phi_i}(x_{\text{clean}} + \delta), \{f_{\phi_i}(x_i^+)\}, \{f_{\phi_i}(x_j^+)\}) , \quad (2)$$

where  $\mathcal{L}$  is the loss function,  $\epsilon$  controls the perturbation budget.

### 3.2. Semantic Contrastive Learning for Perturbation Optimization

Most attacks rely on point-wise cosine similarity, limiting transferability. Inspired by recent efforts that explore contrastive supervision in adversarial optimization [13], we propose a more structured strategy that leverages multiple positive and negative samples to guide the direction of perturbation in a semantically meaningful way.

Let  $z_{\text{adv}} = f_\phi(x_{\text{adv}})$  denote the encoded feature of the adversarial input, and let  $z_i^+ = f_\phi(x_i^+)$  and  $z_j^- = f_\phi(x_j^-)$  be the features of the positive and negative reference samples, respectively. The cosine similarities between  $z_{\text{adv}}$  and the reference features are calculated as:

$$s_i^+ = \frac{\langle \hat{z}_{\text{adv}}, \hat{z}_i^+ \rangle}{\tau}, \quad s_j^- = \frac{\langle \hat{z}_{\text{adv}}, \hat{z}_j^- \rangle}{\tau}, \quad (3)$$

where  $\hat{z} = z/\|z\|_2$  denotes the normalized feature vector and  $\tau$  is a temperature hyperparameter. We compute the normalized log-probability scores for all positive and negative references using the log-softmax operation:

$$\ell_k = s_k - \log \sum_{l=1}^{N_p+N_n} \exp(s_l), \quad \text{for } k = 1, \dots, N_p + N_n, \quad (4)$$

where  $s_k$  denotes the cosine similarity between the  $z_{\text{adv}}$  and the  $k$ -th reference sample (with  $k \leq N_p$  for positive samples and  $k > N_p$  for negative ones), and  $\ell_k$  denotes the normalized log-probability score used for contrastive loss computation. For each input, this log-softmax is computed over its full set of  $N_p + N_n$  references, yielding a vector  $\ell = [\ell_1, \ell_2, \dots, \ell_{N_p+N_n}] \in \mathbb{R}^{N_p+N_n}$ . These scores form the basis for our top- $k$  positive and top- $m$  negative sampling strategy described next.

**Top- $k$  positive contrastive supervision.** Traditional contrastive losses typically treat all positive samples equally, which can dilute gradient signals when some references are noisy, redundant, or only weakly aligned with the target semantics. This is particularly problematic in adversarial optimization, where precise semantic direction is critical. To address this, we introduce a top- $k$  selection mechanism that focuses the optimization on the most semantically relevant references. This prevents overfitting to weak positives while softmax weighting yields smoother, adaptive gradients for better target alignment. In addition, we include a max-based term to ensure that at least one strongly aligned sample dominates the gradient, providing a clear semantic anchor for perturbation learning.

Let  $\ell^+ = [\ell_1^+, \dots, \ell_{N_p}^+]$  denote the log-softmax scores of the  $N_p$  positive reference features. We first select the top- $k$  scores (i.e., the  $k$  largest values) and compute a softmax-

based weighted loss:

$$\mathcal{L}_{\text{pos}}^{\text{top-}k} = - \sum_{i=1}^k w_i \cdot \ell_{[i]}^+, \quad (5)$$

where  $\ell_{[i]}^+$  denotes the  $i$ -th largest entry in  $\ell^+$ , and the weights  $w_i$  are defined by:

$$w_i = \text{softmax}(\text{top-}k(\ell^+)). \quad (6)$$

To complement the soft weighting, we further define a max-based auxiliary loss:

$$\mathcal{L}_{\text{pos}}^{\text{max}} = - \max_{i \in [1, N_p]} \ell_i^+. \quad (7)$$

The final positive loss is a convex combination of the two:

$$\mathcal{L}_{\text{pos}} = \alpha \cdot \mathcal{L}_{\text{pos}}^{\text{top-}k} + (1 - \alpha) \cdot \mathcal{L}_{\text{pos}}^{\text{max}}, \quad (8)$$

where  $\alpha$  is a hyperparameter.

**Top- $m$  negative contrastive supervision.** In contrastive learning, treating all negative samples equally may lead to optimization inefficiency or even instability, especially when many negatives are irrelevant or already well separated from the anchor. This issue is amplified in adversarial settings, where over-penalizing uninformative negatives can dilute gradient signals. To address this, we focus only on the most semantically confusing negative samples—those with high similarity to the adversarial input—and apply targeted suppression.

Let  $\ell^- = [\ell_{N_p+1}^-, \dots, \ell_{N_p+N_n}^-]$  denote the log-softmax scores for the  $N_n$  negative references. We define the negative loss using the top- $m$  entries from this vector:

$$\mathcal{L}_{\text{neg}} = \frac{1}{m} \sum_{j=1}^m \ell_{[j]}^-, \quad (9)$$

where  $\ell_{[j]}^-$  is the  $j$ -th largest entry in  $\ell^-$  (i.e., the  $j$  most confusing negative samples).

This design restricts the loss computation to a small set of hard negatives, thereby concentrating the adversarial signal on separating the adversarial feature from only those features that are semantically close to the original image. (See Appendix 7 for an ablation on negative sampling strategies.)

**Margin-based regularization.** While log-softmax scores guide contrastive optimization, they may fail to reflect true semantic separation in the raw similarity space. We thus introduce a max-margin constraint on cosine similarities to explicitly enforce separation between the most similar positive and the hardest negative reference.

Table 1. Attack success rate (ASR, %) on open-source LVLMs under a fixed perturbation budget of  $\epsilon = \frac{16}{255}$ . We compare our method against five strong targeted black-box transfer attacks under a fair evaluation setup.

| Victim Model              | AttackVLM | AdvDiffVLM | AnyAttack | M-Attack | FOA-Attack | Ours        |
|---------------------------|-----------|------------|-----------|----------|------------|-------------|
| <i>Surrogate Model(s)</i> | ViT-B/32  | Ensemble   | Ensemble  | Ensemble | Ensemble   | Ensemble    |
| InternVL3-14B             | 12.5      | 36.4       | 23.7      | 87.7     | 84.6       | <b>96.7</b> |
| InternVL3-38B             | 6.2       | 14.4       | 11.4      | 54.4     | 54.7       | <b>92.4</b> |
| InternVL3-78B             | 6.0       | 15.1       | 11.0      | 55.6     | 53.6       | <b>94.4</b> |
| Ovis-16B                  | 8.1       | 20.4       | 17.6      | 81.4     | 81.8       | <b>97.3</b> |
| Qwen2.5VL-7B              | 11.8      | 27.2       | 25.6      | 75.5     | 73.9       | <b>93.4</b> |
| Qwen2.5VL-32B             | 12.1      | 28.2       | 25.6      | 72.7     | 72.1       | <b>90.2</b> |
| Qwen2.5VL-72B             | 11.6      | 30.9       | 24.1      | 75.9     | 75.9       | <b>95.0</b> |
| <b>Average</b>            | 9.7       | 24.6       | 19.8      | 71.8     | 70.9       | <b>94.2</b> |

Table 2. Attack success rate (ASR, %) on closed-source LVLMs under a fixed perturbation budget of  $\epsilon = \frac{16}{255}$ . Our method achieves consistently higher transferability across proprietary models, significantly outperforming all baselines.

| Victim Model                   | AttackVLM | AdvDiffVLM | AnyAttack | M-Attack | FOA-Attack | Ours        |
|--------------------------------|-----------|------------|-----------|----------|------------|-------------|
| <i>Surrogate Model(s)</i>      | ViT-B/32  | Ensemble   | Ensemble  | Ensemble | Ensemble   | Ensemble    |
| GPT-4o (2024-11-20)            | 6.5       | 25.1       | 12.7      | 87.6     | 84.6       | <b>95.6</b> |
| Claude-3.7-Sonnet (2025-02-19) | 15.5      | 21.3       | 29.2      | 38.8     | 37.0       | <b>61.1</b> |
| Gemini-2.5-Flash               | 5.9       | 17.8       | 13.4      | 72.5     | 68.7       | <b>92.6</b> |
| <b>Average</b>                 | 9.3       | 21.4       | 18.4      | 66.3     | 63.4       | <b>83.1</b> |

We define the margin loss as:

$$\mathcal{L}_{\text{margin}} = \max \left( 0, \max_j s_j^- - \max_i s_i^+ + \gamma \right), \quad (10)$$

where  $\gamma$  is a margin hyperparameter. This loss ensures that the hardest negative (i.e., the most similar negative) remains less similar to the  $x_{\text{adv}}$  than the most aligned positive reference, by at least a margin of  $\gamma$ .

This structural constraint complements the contrastive supervision by explicitly maintaining semantic separation in the similarity space, thereby improving the stability and robustness of the adversarial optimization.

### 3.3. PCA-Based Subspace Projection for Contrastive Optimization

To stabilize optimization and enhance transferability, we constrain contrastive supervision within a PCA-derived subspace that reduces redundancy and limits surrogate-specific overfitting. While PCA is inherently a linear method and cannot capture nonlinear manifold structures, we find that its top principal components often align with coarse semantic axes in CLIP representations. This, combined with its simplicity and differentiability, makes it a practical and efficient choice for subspace-guided adversarial optimization.

Specifically, given the normalized CLIP features of the adversarial input  $x_{\text{adv}}$  and its reference sets  $\{x_i^+\}$  and  $\{x_j^-\}$ , we denote their representations as  $z_{\text{adv}}, \{z_i^+\}, \{z_j^-\} \in \mathbb{R}^F$ , where  $F$  is the full feature dimensionality. We compute the top- $d$  principal components via PCA:

$$\tilde{z} = U_k^\top z, \quad (11)$$

where  $U_k \in \mathbb{R}^{F \times d}$  contains the top- $d$  principal components. The subspace is updated every  $T$  iterations based on the current features, and contrastive loss is computed within this space. Gradients are projected back to the original space as

$$\nabla_z = U_k \cdot \nabla_{\tilde{z}}. \quad (12)$$

(See Appendix 6 for an AutoEncoder-based variant.)

### 3.4. Overall Loss Function

The final objective combines top- $k$  positive, top- $m$  negative, and margin-based losses, all computed within each surrogate’s PCA subspace. Let  $\mathcal{L}_{\text{pos}}^{(i)}$ ,  $\mathcal{L}_{\text{neg}}^{(i)}$ , and  $\mathcal{L}_{\text{margin}}^{(i)}$  denote the respective loss terms for surrogate  $f_{\phi_i}$ . The total loss is:

$$\mathcal{L}_{\text{total}} = \sum_{i=1}^t \left( \mathcal{L}_{\text{pos}}^{(i)} + \mathcal{L}_{\text{neg}}^{(i)} + \lambda_{\text{margin}} \cdot \mathcal{L}_{\text{margin}}^{(i)} \right), \quad (13)$$



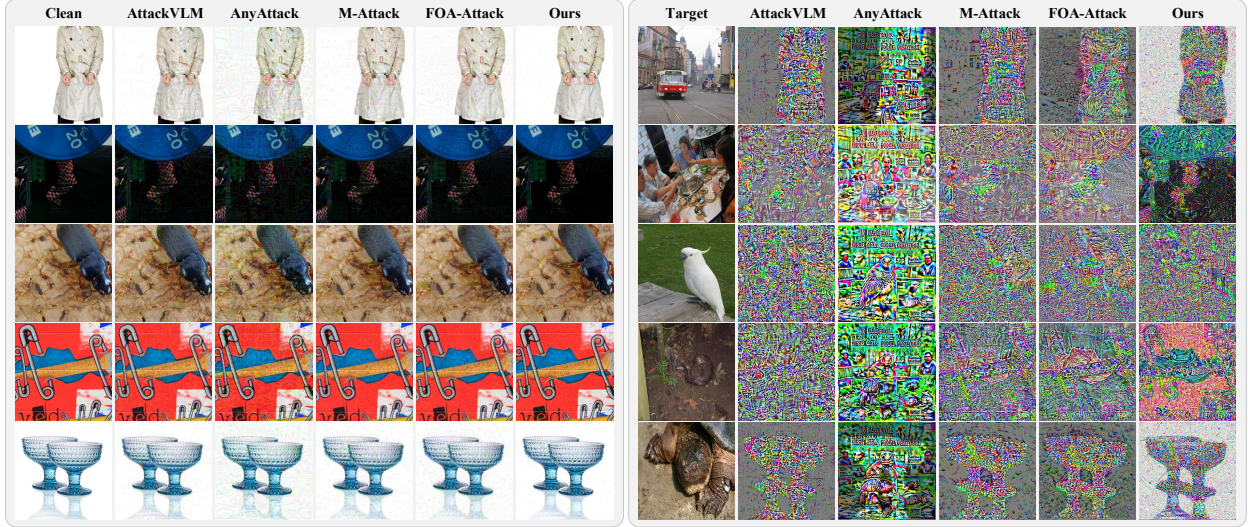


Figure 4. Visual comparison of adversarial examples (left) and corresponding perturbations (right) generated by different methods. Note that the “Target” image shown for our method is only representative—our contrastive supervision is derived from 50 semantically aligned positive samples, not a single target image.

where  $\lambda_{\text{margin}}$  controls the influence of the margin constraint. Gradients are backpropagated through each PCA subspace and projected to update the shared perturbation  $\delta$ .

For optimization, we adopt Momentum Iterative FGSM (MI-PGD) [7], and apply Exponential Moving Average (EMA) smoothing in later stages to stabilize  $\delta$  and improve transferability.

## 4. Experiments

### 4.1. Settings

**Datasets.** Following prior work [8, 13, 19], we use 1,000 clean images ( $299 \times 299$ ) from the NIPS 2017 Adversarial Attack and Defenses Competition dataset. Each image is paired with a target label from a different ImageNet-1K class to define the desired attack semantics. For contrastive supervision, we construct reference sets with 50 positive and 50 negative samples per image, drawn from the ImageNet-1K validation set by matching target and source semantics.

**Surrogate Models.** We use eight CLIP-based surrogates with diverse architectures (ViT, ConvNeXt), resolutions (224–448), and pretraining variants (CLIP, SigLIP, MetaCLIP, DataComp), to ensure robustness across heterogeneous feature spaces. Details in Appendix 1.

**Baselines.** We compare against five recent black-box targeted attack methods: AttackVLM [30], AdvDifVLM [12], AnyAttack [29], M-Attack [19], and FOA-Attack [14]. All baselines are reproduced using their offi-

cial implementations and best-reported hyperparameters to ensure fair comparison. Details in Appendix 4.

**Hyper-parameters.** Unless otherwise specified, we use the following default settings in all experiments:  $\epsilon = \frac{16}{255}$ ,  $N_p = N_n = 50$ ,  $k = m = 10$ ,  $\gamma = 0.2$ ,  $\alpha = 0.8$ ,  $T = 10$ ,  $\lambda_{\text{margin}} = 2.0$ , and PCA dimension  $d = 10$ .

**Evaluation Metrics** We use Attack Success Rate (ASR) as the primary metric, defined as the proportion of adversarial examples whose generated captions are semantically closer to the target meaning than the original. To evaluate this, we adopt the LLM-as-a-Judge paradigm using Qwen2.5-14B-Instruct [24] as a semantic discriminator.

For each adversarial caption  $c_{\text{adv}}$ , the LLM is prompted to determine whether its semantics align more with the original or the target description. To mitigate known biases in LLM judgment, we apply:

- **Position Balancing:** The order of the two candidate semantic descriptions (Options A/B) is randomized, and each sample is evaluated twice. Success is recorded only if the target meaning is preferred in both versions.
- **Prompt Standardization:** Prompts follow a fixed structure and consistent wording across all samples.
- **Deterministic Decoding:** Sampling is disabled to ensure reproducible outputs. (do\_sample=False)
- **Human Verification:** A 100-sample subset was manually reviewed, achieving 91.0% agreement with LLM decisions and a Cohen’s  $\kappa$  of 0.86.

This evaluation protocol ensures that ASR reflects gen-

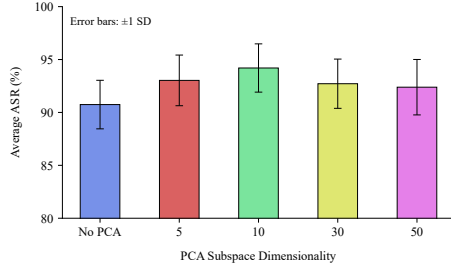


Figure 5. Average ASR across seven open-source LVLMs under different PCA subspace dimensionalities.

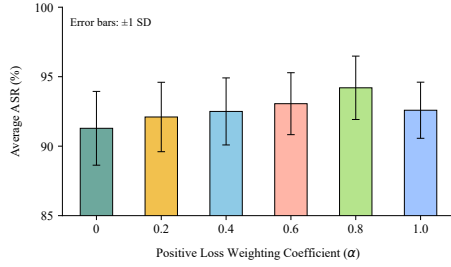


Figure 6. Average ASR across seven open-source LVLMs under different positive loss weighting coefficients ( $\alpha$ ).

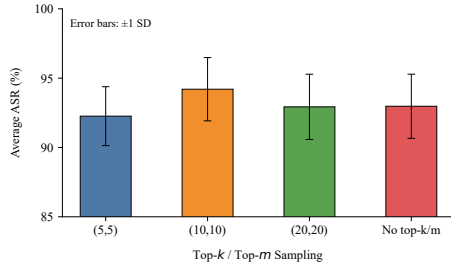


Figure 7. Average ASR across seven open-source LVLMs under different Top- $k$  and Top- $m$  sampling strategies.

uine semantic alignment rather than artifacts of prompt design or decoding randomness. Full prompt templates are provided in Appendix 2.

## 4.2. Results

**Comparison with Baselines on Open-Source LVLMs.** Under a unified evaluation setting with shared inputs, target semantics, and LLM-as-a-judge paradigm, our method consistently outperforms all baselines across seven open-source LVLMs ( $p < 0.05$ ), as shown in Table 1.

While several baselines achieve moderate performance on smaller models (e.g., InternVL3-14B), their ASR drops significantly as model size increases. For instance, M-Attack drops from 87.7% on InternVL3-14B to 54.4% and

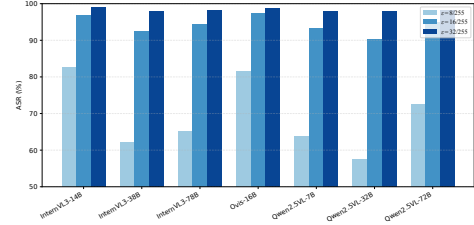


Figure 8. Effect of perturbation budget  $\epsilon$  on average ASR across seven open-source LVLMs.

55.6% on 38B and 78B, respectively; FOA-Attack exhibits a similar decline (84.6%  $\rightarrow$  54.7% / 53.6%). In contrast, our method maintains high ASR above 92% across all InternVL variants, indicating strong robustness to model scaling.

A similar trend is observed on the Qwen2.5VL series: while M-Attack fluctuates between 72.7% and 75.9%, our method consistently achieves over 90% ASR, reaching 95% on Qwen2.5VL-72B. Notably, the maximum performance gap between our method and the strongest baseline reaches 40.8 percentage points on InternVL3-78B, underscoring the advantage of our semantic-guided optimization.

### Comparison with Baselines on Closed-Source LVLMs

We further evaluate transferability on three proprietary LVLMs: Claude-3.7-Sonnet, Gemini-2.5-Flash, and GPT-4o (Table 2). Our method achieves the highest ASR on all targets.

While M-Attack and FOA-Attack remain competitive on GPT-4o (87.6% and 84.6%), their ASR drops sharply on Claude-3.7 (38.8% and 37.0%). In contrast, our method yields stable and superior ASR: 95.6% on GPT-4o, 92.6% on Gemini-2.5, and 61.1% on Claude-3.7.

Notably, the performance gap between our method and the next-best baseline exceeds 22.3 percentage points on Claude 3.7 and 20.1 percentage points on Gemini 2.5, highlighting the robustness of our semantic-guided optimization under real-world black-box constraints.

### 4.3. Ablation Studies

**Effect of PCA Subspace Dimensionality.** Fig. 5 shows that removing PCA projection significantly reduces ASR (to 90.5% on average), confirming that direct optimization in full feature space introduces noise and harms transferability. A subspace dimension of 10 achieves the highest ASR (96.7%), while both smaller (5D) and larger (30D, 50D) dimensions degrade performance, suggesting a balance between compactness and expressiveness is critical.

**Effect of Positive Loss Weighting Coefficient  $\alpha$ .** Varying the positive loss coefficient  $\alpha$  from 0 to 1 (Fig. 6) shows that  $\alpha = 0.8$  yields the best ASR. This supports combining

top- $k$  weighting and max-based supervision for more stable optimization.

**Effect of Top- $k$  and Top- $m$  Sampling in Contrastive Supervision.** Fig. 7 compares sampling strategies. The (10,10) configuration consistently outperforms others, indicating an effective balance between relevance and diversity, while larger or no filtering leads to noisier supervision.

**Effect of Perturbation Budget  $\epsilon$ .** Increasing  $\epsilon$  from 8/255 to 32/255 improves ASR up to 98.2% (Fig. 8), confirming that a larger perturbation budget enhances black-box transferability.

#### 4.4. Visualization

Compared to other methods, our perturbations exhibit three notable characteristics:

**Semantic Localization:** The learned perturbations exhibit strong alignment with semantically salient regions (e.g., subject foreground), rather than spreading uniformly or randomly—improving target alignment while preserving overall image quality. This effect is most apparent in Rows 1, 4, and 5.

**Semantic-Rich Guidance:** Our method leverages a positive set of 50 semantically related images, enabling richer supervision and smoother perturbation structures than single-target baselines.

**Visual Consistency:** Instead of relying on arbitrary noise, our method exploits all semantically informative features available in the clean image to steer the output toward the target semantics. This results in adversarial examples that are visually aligned with the original input while exhibiting coherent perturbation patterns.

These results support the effectiveness of our semantic-aware, subspace-guided strategy in producing transferable and interpretable adversarial examples.

### 5. Discussion

#### 5.1. Why Constraining Optimization to a PCA Subspace Enhances Transferability?

Constraining adversarial optimization to a PCA subspace acts as an explicit capacity-control mechanism that suppresses surrogate-specific overfitting in high-dimensional CLIP/SigLIP feature spaces. These representations are known to exhibit strong spectral anisotropy: only a small fraction of principal directions encode cross-model semantic regularities, whereas the majority of low-variance directions capture encoder-dependent artifacts or dataset-specific biases. Optimizing perturbations in the full feature space therefore provides excessive degrees of freedom, allowing gradients to drift into brittle, low-energy directions that yield surrogate-specific alignment but fail to generalize to

unseen LVLMs. This phenomenon mirrors the behavioral gap observed in over-parameterized predictors trained on limited data, where many parameters are weakly supported by the training signal. By projecting features onto the principal subspace with the highest sample variance, PCA effectively matches the optimization capacity to the amount of semantic evidence available in the reference sets. As a result, perturbation updates are forced to follow dominant, semantically meaningful variations that are shared across architectures, leading to substantially improved black-box transferability. (See Appendix 3)

#### 5.2. Why Contrastive Supervision Produces More Transferable Targeted Attacks?

Contrastive supervision improves transferability through a mechanism fundamentally different from PCA: it reshapes the optimization objective to enforce structured semantic alignment rather than merely maximizing point-wise similarity. Traditional attacks rely on optimizing a single cosine similarity, which gives an under-constrained objective that is easily satisfied by exploiting spurious directions in the surrogate encoder. In contrast, contrastive learning anchors the adversarial feature to a distribution of positive and negative samples, requiring the perturbation to satisfy a set of mutual relationships rather than a single alignment target. This multi-reference structure stabilizes geometric relationships, mitigates mode collapse, and forces perturbations to move toward semantic regions consistently associated with the target concept. Furthermore, the top- $k$  positive and top- $m$  hard negative sampling strengthens gradients along directions that are both discriminative and semantically informative, while suppressing weak or contradictory signals. This distributed supervision prevents the attack from overfitting to idiosyncratic features of any specific surrogate, thereby promoting perturbations that generalize more effectively across LVLMs with heterogeneous architectures, modalities, and pretraining corpora.

### 6. Conclusion

In this paper, we propose **VCP-Attack**, a targeted adversarial attack framework that achieves **state-of-the-art** transferability on LVLMs via PCA-guided subspace projection and contrastive supervision. Our results expose critical vulnerabilities in both open-source and proprietary models (e.g., GPT-4o, Claude, Gemini), where carefully crafted perturbations lead to consistent misalignment of model outputs. These findings underscore the urgency of developing robust defense mechanisms for secure LVLM deployment. Although our experiments focus specifically on image captioning, this fundamental multimodal task serves as a representative setting for evaluating black-box adversarial transferability. While promising, further validation on other vision-language tasks remains a direction for future work.



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